

soft-margin SVM: find w, b that minimise

margin error

$$\frac{\|w\|}{2}$$

small when the
margin is wide

+

C

knob

×

classification error

$$\sum_{i=1}^n \xi_i$$

how far points sit on
the wrong side of
their own hyperplane

large C: avoid mistakes, accept a narrow margin
small C: keep the margin wide, accept some mistakes